Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **backend/full-stack systems** and **QA automation workflows** using **Java**, **Python**, SQL, and Linux-based tooling, focused on **test reliability**, **CI/CD validation**, and production debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Designed and validated backend data workflows across **Azure SQL/PostgreSQL** for 500+ users, reducing dashboard latency from **1.2s to 350ms**.
- Implemented release checks and reconciliation-driven validation for 5K+ monthly jobs, improving **reliability** and defect prevention in production environments.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed and tested a Flask + React platform handling **10K+ daily requests** with modular APIs and instructor-facing quality checks.
- Added concurrent processing and CI/CD controls with GitHub Actions, reducing turnaround time by **60%** while maintaining stable API behavior under peak loads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built Python experimentation/simulation pipelines processing **100K+ trajectory logs** with reproducible test runs and structured outputs.
- Documented validation outputs and debugging findings to improve issue triage and cross-team troubleshooting workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Developed and maintained **20+ unit/integration test suites** using Jenkins, PyTest, and Mockito across 3 OS environments, strengthening release quality gates.
- Troubleshot API/ETL defects with OOP, AWS Glue, and S3, improving response time by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built REST API and WebSocket modules supporting 8+ workflows and 2M+ synchronized records with clear operational logging and data checks.
- Automated deployment/testing with Jenkins and Docker, reducing deployment time by **50%** and improving release consistency through change cycles.

Skills

Languages: **Java**, **Python**, SQL, JavaScript/TypeScript, C++
QA & Automation: Unit testing, integration testing, Selenium familiarity, CI/CD, test case design, defect triage
Backend & APIs: Flask, FastAPI, Node.js, Express.js, REST APIs, JSON, OOP
Scripting & Platforms: Linux/Unix command-line workflow, shell scripting, Jenkins, GitHub Actions, Git
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL
Engineering Practices: SDLC, Agile/Scrum, debugging, production support, technical documentation

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built an AI-powered SQL workspace for 10+ databases with schema-aware query validation, structured summaries, and reproducible execution workflows.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built secure web experiences for 50K+ users and 1M+ transactions with real-time updates, scalable deployment, and reliable data workflows.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed API-driven healthcare operations with monitoring-backed workflows, supporting 100+ daily appointments and 2x faster record retrieval.